VAIBHAV LALWANI

APPLIED AI & PYTHON BACKEND ENGINEER

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Portfolio | GitHub | LinkedIn

Professional Experience

Full-Stack / AI Engineer | Meta Solution Technologies Apr–May 2026

- Built an AI-guided admissions workflow using Next.js, React, TypeScript and Python-backed REST APIs; integrated Supabase authentication and PostgreSQL/pgvector retrieval.
- Implemented document-processing and email workflows using Redis/BullMQ; handled validation, failure states and multilingual interfaces.

Software Engineer (Part-time) | Recruit Pilot Jan–Apr 2026

- Delivered React/TypeScript features and REST API integrations for a recruitment product, participating in code review and weekly release cycles.

Coding Instructor (training) | RoboCode UK Aug 2026–Present

- Completed instructor induction; preparing Python and robotics lesson exercises and troubleshooting examples.

Front-End Developer Intern | Unified Mentor Mar–May 2025

- Built React features for a learner-facing platform, including lessons, progression views and REST-connected dashboards.

Selected Engineering Projects

HydraSentry | AI-agent memory security SaaS (FastAPI, Next.js) Source | Demo

- Built a multi-tenant FastAPI backend with Supabase magic-link auth, per-user API keys, rate limiting and an MCP gateway that blocks poisoned-memory actions; 237 pytest tests pass, CI on every push.

Lacuna | Provenance-first memory for AI agents Source | Demo

- Built temporal retrieval across web, CLI and MCP surfaces that abstains when evidence conflicts; 2,348 Vitest checks pass; includes a LongMemEval benchmark harness with recorded runs.

QueueProof | Cross-source evidence workspace Source | Demo

- Retrieval pipeline with claim-level citations, degraded-source handling and an approval boundary for external writes; 654 Vitest tests pass; evaluation fixtures, grading scripts and measured results are published in the repo.

Kodro | Offline-first robotics learning studio (Python) Source | Demo

- Pygame/Pymunk simulation app with Pydantic-validated models, an enforced 85% coverage gate and reproducible scenario evaluations committed as data.

Selected Achievements

HydraDB x Connectors Winner (QueueProof) | SharedOS 2026 Judges' Pick (Yuzu) | HydraDB WikiThon Finalist (QyntraWiki Nexus) | Product Hunt #24 of the day (Kodro)

Education

MSc Advanced Data Science & Artificial Intelligence | University of Liverpool Expected Jan 2027

BCA | Christ University, Bengaluru 2022–2025

Certifications

Artificial Intelligence on Microsoft Azure | Azure Machine Learning | Google Introduction to AI

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
APIs and backend: FastAPI, REST, Node.js, PostgreSQL, Supabase, Redis, BullMQ, Docker
AI engineering: LLM APIs, RAG and retrieval evaluation, MCP servers, structured output validation with fallbacks, prompt/injection red-teaming
Experiment practice: benchmark fixtures, logged and reproducible runs, error-handling and timeout paths, cost-conscious local-model evaluation, Git, GitHub Actions, pytest, Vitest